23 January 1970
Superseding
MIL-S-55226(EL)
20 March 1963

MILITARY SPECIFICATION

SIMULATOR, ANTENNA POSITION SM-154( )/MPQ-4A

1. SCOPE

1.1 This specification covers one type of antenna simulator designated Simulator, Antenna Position, SM-154( )/MPQ-4A which furnishes azimuth, elevation, and scanner pulses identical to those normally provided by the Radar Set AN/MPQ-4A system antenna. (See 6.4)

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of the invitation for bids or request for proposal form a part of the specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

QQ-S-571	Solder; Tin Alloy, Lead-Tin Alloy, and Lead Alloy
PPP-B-636	Box, Fiberboard
PPP-B-640	Box, Fiberboard, Corrugated, Triple-Wall
PPP-F-320	Fiberboard, Currugated and Solid, Sheet Stock. Container Grades and Cut Shape
PPP-S-760	Strapping, Nonmetallic, and Connectors
PPP-T-97	Tape, Pressure-Sensitive Adhesive, Filament Reinforced

MILITARY

MIL-P-116	Preservation, Methods of
MIL-V-173	Varnish, Moisture-and-Fungus-Resistant
MIL-W-6858	Welding, Resistance, Aluminum, Magnesium, Non-Hardening Steels or Alloys, Spot
MIL-B-7883	Brazing of Steels, Copper, Copper Alloys, Nickle Alloys, Aluminum and Aluminum Alloys

FSC 6625

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:39Z
Check the source to verify that this is the current version before use.

MIL-M-10578	Metal Conditioner and Rust Remover-Phosphoric Acid Base
MIL-M-13231	Marking of Electronic Items
MIL-F-14072	Finishes for Ground Signal Equipment
MIL-F-14256	Flux, Soldering, Liquid-Rosin Base
MIL-T-22361	Thread Compound; Antiseize,
MIL-B-43291	Box, Fiberboard, Corrugated, Double-Wall, Weather Resistant

STANDARDS

MILITARY

MIL-STD-22	Welded-Joint Designs
MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-461	Electromagnetic Interference Characteristics Requirements for Electronic Equipment
MIL-STD-810	Environmental Test Methods

DRAWINGS

ELECTRONIC COMMAND

SC-B-61578	Grounding to Chassis
SC-DL-432500	Simulator, Antenna Position, SM-154( )/MPQ-4A

(Copies of documents required by contractors, in connection with specific procurement functions, should be obtained from the procuring activity or as directed by the Contracting Officer. Both the title and number or symbol should be stipulated when requesting copies.)

3. REQUIREMENTS

3.1 Description. - Simulator Antenna Position, SM-154( )/MPQ-4A, is used when bench servicing Indicator, Azimuth and Range IP-375( )/MPQ-4A and Computer, Radar Data, CP-319( )/MPQ-4A. It also supplies the trigger pulse for servicing Amplifier, Trigger Pulse, AM-1537( )/MPQ-4A and Control, Power Supply C-2014( )/MPQ-4A. The following outputs are supplied:

Deadtime pulses
Azimuth strobe pulses
Trigger Pulses
Azimuth synchro data

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:39Z
Check the source to verify that this is the current version before use.

Elevation synchro data
Azimuth strobe synchro data

3.2 First Articles. - The contractor shall furnish "First Articles" for approval, if required by the Invitation for Bids and the contract. (See 4.3 and 6.1e(1)).

3.3 Physical construction. - Physical construction of the equipment shall be in accordance with the drawings and documents listed in SC-DL-432500.

3.4 Processes and materials. -

3.4.1 Finish. - The equipment shall be finished in accordance with Specification MIL-F-14072 and the equipment drawings. (see 4.4)

3.4.2 Marking. - Marking shall conform to Specification MIL-M-13231 (see 4.4). Front panel marking shall be group 1 as described in that specification.

3.4.2.1 Serial numbers. - Each unit shall be serial numbered.

3.4.3 Cleaning. -

3.4.3.1 Parts. - After fabrication, parts shall be cleaned in accordance with good commercial practice, or as specified in an applicable document. Cleaning processes shall have no deleterious effect. Corrosive material shall be removed completely before the parts are mounted on chassis, panels, etc.

3.4.3.2 Units. - After assembly, units shall be cleaned thoroughly and shall be free from particles of solder, flux, and other foreign material. In addition, when necessary, such cleaning shall also be performed before final assembly of the units.

3.4.4 Brazing. - Brazing of steels, copper, copper alloys, and nickel alloys shall conform to Specification MIL-B-7883.

3.4.5 Soldering. -

3.4.5.1 Solder. - Solder shall conform to Specification QQ-S-571. For soldering electrical connections, composition SN60, type AR or S, shall be used for general purposes; and composition SN62 or SN63, type AR or S, for special purposes. When type S solder is used for soldering electrical connections, the flux shall conform to Specification MIL-F-14256.

3.4.5.2 Acid or acid salts. - No acid or acid salts shall be used in preparation for or during soldering; however, exception is permitted for preliminary tinning of electrical connections and for tinning or soldering of mechanical joints not used to complete electrical circuits, but in no case shall acid or acid salts be used where they can come in contact with insulation material. Where acid or acid salts are used, as permitted above, they shall be completely neutralized and removed immediately after use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:39Z
Check the source to verify that this is the current version before use.

3.4.5.3 Process. - There shall be no sharp points or rough surfaces resulting from insufficient heating. The solder shall feather out to a thin edge, indicating proper flowing and wetting actions, and shall not be crystallized, overheated, or underheated. The minimum necessary amount of flux and solder shall be used for electrical connections. Any means employed to remove an unavoidable excess of flux shall not incur the risk of loose particles of flux, brush bristles, or other foreign material remaining in the equipment; flux being spread over a larger area; or damage to the equipment. Insulation material that has been subjected to heating during the soldering operation shall be undamaged and parts fastened thereto shall not have become loosened.

3.4.6 Welding. - Wherever practicable, welded joints shall conform to Standard MIL-STD-22 and shall be such that grinding on the finished weld will be unneccessary. Spot, stitch, and seam welds shall conform to Specification MIL-W-6858.

3.4.6.1 Cleaning prior to welding. - Surfaces to be welded shall be cleaned in accordance with good commercial practice and shall be free from rust, scale, paint, grease, and other foreign material.

3.4.6.2 Process. - Preheating shall be employed where distortion is likely to result from welding. Welds shall have thorough penetration and good fusion and shall be free from scabs, blisters, abnormal pack marks, cracks, voids, slag inclusions, and other harmful defects. Where undesirable internal stresses are likely to result from welding, welded items shall be stress-relieved. Inert-gas-shielded arc welding shall be used, when practicable, for welding of aluminum, magnesium, or stainless steel.

3.4.6.3 Cleaning after welding. - Welded assemblies shall be cleaned to remove rust, scale, oxidation products, and excess flux by sandblasting, wire brushing, or other suitable means. Prior to painting, steel parts that have been arc welded or acetylene welded shall, in addition, be subjected to vat passivation or a phosphoric acid etch in accordance with Specification MIL-M-10578. Acid used for cleaning shall be completely neutralized and removed.

3.4.7 Riveting. - Riveting shall not be used to fasten maintenance parts such as capacitors, resistors, transformers, inductors, and tube sockets. After riveting, the joints shall be tight, the joined parts shall be undamaged, and the rivet heads shall be properly seated and tight against their bearing surfaces.

3.4.8 Wiring protection. - Wires shall not be bent sharply where they enter insulation material.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:39Z
Check the source to verify that this is the current version before use.

3.4.8.1 Insulating sleeving. - Insulating sleeving shall not be used unless specified for a particular application or specifically approved by the contracting officer. Twine or tape shall not be used for securing wire and cable. (NOTE: When shielding on wire or cable is unprotected by an outer insulation, adequate support is necessary to prevent the shielding from coming in contact with exposed terminals or conductors.)

3.4.9 Splicing. - Wires in a continuous run between two terminals shall not be spliced.

3.4.10 Terminals, fixed. - Fixed terminals (including binding posts) shall properly support the wires connected thereto and shall not turn or become loose when the items in which the terminals are used are subjected to specified service conditions.

3.4.11 Connections. - Before being soldered to terminal lugs or fixed terminals, wires shall be mechanically secured so that the connections are not dependent for strength on solder alone. Where practicable, wires soldered to fixed terminals shall be looped at least one-half turn but not more than three-quarters turn around the terminal before soldering. Bared ends of wire leads to be terminated in solder-type terminal lugs shall be tinned, silver plated, or lead-alloy coated. Electrical connections shall not be made by clamping between a metallic and nonmetallic material. Fraying of textile ends of wires shall be prevented mechanically or by application of varnish conforming to Specification MIL-V-173. No varnish, lacquer, inspection paint, or other coating shall be applied to completed electrical connections except that these connections need not be masked during overall treatment with transparent protective coating materials.

3.4.12 Grounding to chassis. - Grounding to chassis shall conform to Drawing SC-B-61578.

3.4.13 Torque-measuring devices. - Threaded parts of steel shall be tightened to the torque valves shown in Table I. Threaded parts of other size or material shall be tightened to a suitable design value established by the contractor for the particular application. For critically stressed applications, the torque for tightening threaded parts shall be controlled by use of torque measuring or controlling devices.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:39Z
Check the source to verify that this is the current version before use.

Table I - Torque for tightening steel threaded parts

Thread	Torque (in-lb)
6-32	7 to 9
8-32	14 to 18
10-24	19 to 23
10-32	25 to 30
12-24	32 to 39

3.4.14 Anti-seize compound. - Anti-seize compound, conforming to MIL-T-22361, shall be used to prevent seizing of threads where the threaded surface is aluminum or magnesium.

3.4.15 Plastics. - Where not machined, plastic material and parts shall have the original smooth or polished surfaces. Surfaces that have been sawed, cut, punched, or otherwise machined shall be as smooth as practicable in accordance with good manufacturing practice for the intended application.

3.5 Interchangeability. - Like units, assemblies, subassemblies, and replaceable parts shall be physically and functionally interchangeable without modification of such items or of the equipment. (See 4.6) Individual items shall not be hand-picked for fit or performance.

3.6 Preconditioning. - The equipment shall be capable of meeting the requirements herein, without subsequent processing, after subjection to the bounce preconditioning of 4.7 (Also see 4.5).

3.7 Service conditions. - The equipment shall meet the following service conditions.

3.7.1 Operation. - Continuous use for a period of five years, under the worldwide environmental conditions specified in 3.7.2 through 3.7.9, with a duty cycle of 16 hours on and eight hours off and with no more than normal maintenance and replacement of parts. Where the contractor is required to make a selection of parts, materials processes, construction methods, etc. he shall be guided by the following conditions. Approval of the First Article sample (See 3.2) shall be considered as compliance with this requirement.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:39Z
Check the source to verify that this is the current version before use.

3.7.2 Temperature. -

Operating. - +125°F (51.7°C) to 32°F (0°C)

Non operating. - +185°F (85°C) to -76°F (-60°C)

3.7.2.1 Low. - During testing, as specified in 4.8.1, the equipment shall meet the following requirements of SM-B-432731: (listed in SC-DL-432500)

Paragraph	Measurement	Performance
4.3.6	(Start pulse, stop pulse and) (strobe pulse)	See SM-B-432731
	Pulse width of each	
	Overshoot of each.	
4.3.12	Delay time	
4.3.8	Time between pulses	
4.3.13	Gain	
4.4.3	Trigger pulse generator	
	Amplitude	
	Pulse width	
4.4.5	Pulse frequency	
4.4.7	Pulse Amplitude	

3.7.2.2 High. - During testing as specified in 4.8.1, the equipment shall meet the same requirements as listed in 3.7.2.1 above.

NOTE: After conclusion of the temperature cycling the equipment shall meet the complete requirements of SM-B-432731.

3.7.3 Humidity (Moisture-resistance). - During testing, as specified in 4.8.2, the equipment shall meet the requirements of 3.7.2.1, above. After completion of the humidity cycling the equipment shall meet the complete requirements of SM-B-432731.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:39Z
Check the source to verify that this is the current version before use.

3.7.4 Altitude (elevation). - During testing, as specified in 4.8.3, the equipment shall meet the requirements of 3.7.2.1 above. After completion of the altitude cycling the equipment shall meet the complete requirements of SM-B-432731.

3.7.5 Vibration. - The equipment shall be vibrated in accordance with 4.8.4. During this vibration cycling the amplitude of vibration of the mounted equipment, or any part thereof, shall not exceed three times the amplitude of the applied vibration.

3.7.6 Bench handling. - The equipment shall meet all specified performance requirements, herein, and shall show no evidence of damage, after subjection to the bench handling test of 4.8.5.

3.7.7 Drop. - The case shall meet the requirements of the drop test portion of Drawing SM-B-432737, after subjection to the drop test of 4.8.6 herein.

3.7.8 Air Seal test. - The case shall pass the air seal test of 4.8.7.

3.7.9 Fungus. - The equipment shall show no evidence of viable fungus or corrosion when subjected to the test specified in 4.8.8. Corrosion is any visible degredation that can be attributed to flaky, pitted, blistered, or otherwise loosened finish or metal surface.

3.8 Performance requirements. -

3.8.1 Power input. - The simulator shall operate at full specified performance with the following power inputs:

a. $115\ ^{+15}_{-5}$ volts (v), alternating current (ac), 400 cycles per second (cps) $\pm$ five percent, three phase (Wye configuration).

b. $115\ ^{+15}_{-5}$ v ac, 400 cps $\pm$ five percent, single phase.

c. + 220 $\pm$ 5 v, direct current (dc).

d. -220 $\pm$ 5 v dc.

3.8.2 Antenna pulse simulator (Blocks 2100 and 2800). - The simulated antenna pulses shall be in accordance with the requirements listed in SM-B-432731.

3.8.3 Wiring of Block 2700. - Block 2700 shall be wired in accordance with SM-E-432652 (listed in SC-DL-432500).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:39Z
Check the source to verify that this is the current version before use.

3.8.3.1 Blown fuse indicators. - The blown fuse indicators shall glow when a fuse opens, or is removed from its circuit.

3.8.4 Position synchros. - The position synchros shall operate in accordance with the following:

a. Azimuth position (Block 2300) per SM-B-432661 (listed in SC-DL-432500)

b. Elevation position (Block 2400), per SM-B-432662 (listed in SC-DL-432500)

c. Azimuth Marker (Block 2500), per SM-B-432749 (listed in SC-DL-432500)

3.8.5 Radio interference suppression. - The equipment shall conform to Specification MIL-STD-461, Table A - II - MIL-I-11748, Class IC. (See 4.9.5)

3.9 Technical literature tools, and running spare parts. - Technical literature tools, and running spare parts shall be furnished, if specified in the contract. Running spare parts shall be identical to corresponding parts in the equipment furnished on the order. (See 6.1h and i)

3.10 Workmanship. - The equipment shall be manufactured and assembled in accordance with the applicable portions of the following paragraphs herein:

3.3	Physical construction	3.4.8	Wiring protection
3.4.2	Marking	3.4.9	Splicing
3.4.3	Cleaning	3.4.10	Terminals, fixed
3.4.5	Soldering		
3.4.6	Welding	3.4.11	Connections
3.4.7	Riveting	3.4.12	Grounding to chassis
		3.4.15	Plastics

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:39Z
Check the source to verify that this is the current version before use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspection. - Inspection shall be classified as follows:

a. First Article inspection (does not include preparation for delivery). (See 4.3)

b. Inspection covered by subsidiary documents. (See 4.4)

c. Quality conformance inspection.

(1) Quality conformance inspection of equipment before preparation for delivery. (See 4.5)

(2) Quality conformance inspection of preparation for delivery. (See 4.12)

4.3 First Article inspection. - This inspection shall consist of the inspection specified in the subsidiary documents covering the items listed in 4.4, and the inspection specified for group A, group B, and group C (See Tables III, IV, and para. 4.5.3, respectively). The preproduction inspection shall be performed in the order shown in Table II according to the number of samples specified in the contract or order.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:39Z
Check the source to verify that this is the current version before use.

Table II - Order of First Article inspection

Inspection (Note A) (Also See 4.10)	Test Para	Number of samples					
		One	Two		Three		
			Unit 1	Unit 2	Unit 1	Unit 2	Unit 3
Non-damaging (Note B)							
Temperature							
Low	4.8.1	1	1		1		
High	4.8.1	2	2		2		
Altitude (Elevation)	4.8.3	3		1		1	
Vibration	4.8.4	4		2		2	
Air Seal Test (case only) (Note D)	4.8.7	6		3		4	
Radio Interference Suppression (Note C)	4.9.5	8	4		4		
Potentially damaging							
Humidity (moisture-resistance)	4.8.2	5	3		3		
Shock, drop (Case only)	4.8.6	7		4		3	
Fungus	4.8.8	9	5				1

Note A: Other First Article inspection may precede, follow, or be interspersed between the non-damaging tests.

Note B: The order on non-damaging tests and the choice of which units are subjected to the non-damaging tests may be varied if convenient.

Note C: Performed on Preproduction Sample only.

Note D: Performed on Preproduction and Group A tests only.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:39Z
Check the source to verify that this is the current version before use.

4.4 Inspection covered by subsidiary documents. - The following shall be inspected under the applicable subsidiary documents as part of the inspection of equipment before preparation for delivery:

Item	Where required
Finish	3.4.1
Marking	3.4.2

4.5 Quality conformance inspection of equipment before preparation for delivery. - Each unit of each lot of equipment shall be subjected to bounce preconditioning (See 3.6) prior to inspection. The contractor shall perform the inspection specified in 4.4 and 4.5.1 through 4.5.4. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. Failure to meet any requirement as a result of bounce preconditioning shall be classified as a defect and recorded as such. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition the Government - - at its discretion - - may perform all or any part of the specified inspection, to verify the contractor's compliance with specified requirements. (See 6.5) Test equipment for Government verification inspection shall be made available by the contractor.

4.5.1 Group A inspection. - Each unit of each lot of equipment shall be inspected for conformance to all the inspection and test requirements of Table III. Each lot will be subject to verification, utilizing the procedures of MIL-STD-105 using the general inspection levels and the AQL's indicated in Table III. Group A inspection shall be performed in any order which is satisfactory to the Government, except that the air-seal test (4.8.7) shall be last. (See 6.3)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:39Z
Check the source to verify that this is the current version before use.

Table III - Group A inspection

Inspection	Req. Para.	Insp. Para.	AQL Major	AQL Minor
Visual and Mechanical	3.10	4.11	2.5 dphu	10.0 dphu
Air Seal	3.7.8	4.8.7	1.0%	* *
Operational				
Continuity	3.8.3	4.9.3	1.0%	
Blown fuse indicator	3.8.3.1	4.9.3.1	1.0%	
Positive Synchros	3.8.4	4.9.4		* *
Azimuth	a	4.9.4	2.5 dphu for the group	
Elevation	b	4.9.4		
Azimuth Modes	c	4.9.4		
Antenna pulse simulator	3.8.2	4.9.2	1.0%	

* dphu = defects per hundred units

* * All defects are considered to be major only.

4.5.2 Group B inspection. - This inspection, including sampling, shall conform to Table IV and to special procedures for small - sample inspection of MIL-STD-105. Group B inspection shall normally be performed on inspection lots that have passed Group A inspection and on samples selected from units that have been subjected to and met the Group A inspection.

Table IV - Group B inspection

Inspection	Req. Para.	Insp. Para.	AQL (See 4.5.2.1)
Shock, bench handling	3.7.6	4.8.5	4.0%
Interchangeability	3.5	4.6	6.5%

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:39Z
Check the source to verify that this is the current version before use.

4.5.2.1 Group B sampling plans. - The Group B sampling plans, for the AQL's listed in Table IV, shall be as follows:

AQL	Inspection level
4.0%	S-4
6.5%	S-4

4.5.2.2 Order of inspection within Group B. - Group B inspection shall be performed in the order proposed by the contractor and accepted by the Government.

4.5.3 Group C inspection. - This inspection shall consist of the tests specified in Table II and shall be performed on sample units that have been subjected to and met Group A and Group B inspection. Each sample unit used for the humidity test need meet only those environmental Group B inspections indicated for that unit in Table II. Sample units shall be selected in accordance with paragraph 4.5.3.1. During the Air Seal test the equipment shall be installed in the case.

4.5.3.1 Sampling for inspection. - Two (2) sample units shall be selected at random for all inspections in Table II from every fifty (50) and fraction thereof produced. The first samples selected shall be at the start of the contract from the first quality conformance inspection production lot. (See 4.5.3)

4.5.3.2 Noncompliance. - The contractor shall immediately report in writing each Group C failure occurrence, including details of the failure and characteristics affected. The contractor shall immediately investigate the cause of failure and further report the results of investigation and details of the proposed corrective action on (i) the process and materials, as applicable, and, (ii) all units of product which were manufactured under the same conditions and which the Government considers subject to the same failure. Reports shall be forwarded to the responsible technical activity designated in the contract through the Quality Assurance Representative. After corrective action has been taken, additional sample units shall be subjected to Group C inspection (all inspections, or the inspections which the sample failed, at the option of the Government) and Groups A and B inspection may be reinstituted; however, final acceptance and shipment will be withheld until the Group C reinspection results have shown that the corrective action was effective.

4.5.4 Reinspection of Conforming Group B and Group C Sample Units. - Unless otherwise specified, sample units which have been subjected to and passed both Group B and Group C inspection may be accepted on contract or order provided all damage is repaired and the sample units are resubjected to and pass Group A inspection.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:39Z
Check the source to verify that this is the current version before use.

4.6 Inspection for dimensional interchangeability. - The dimensions listed below shall be gaged or measured to determine conformance to the physical interchangeability requirement of 3.5. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

a. External and internal dimensions of cases, covers and insertable assemblies, when such dimensions affect mating of parts.

b. Dimensions of cavities, when such dimensions affect insertion of items.

c. Location of hinges and fasteners on separable parts or assemblies which must mate, such as cases, covers, and mountings.

d. Location of connectors, locking pins, fasteners, slides, and mountings which receive mating parts of plug-in assemblies and major units; and location of the mating parts on the plug-in assembly or major unit.

e. Size and form of special threads.

4.7 Bounce preconditioning. - The unit shall be placed in its normal operating position on the table of the Package Tester, Type 1000-SC, as made by the L.A.B. Corporation, Skaneateles, New York, or equal. The package tester, shafts in phase, shall have a speed such that it is just possible to insert a 1/32-inch-thick strip of material under one corner or edge of the unit to a distance of 3 inches as the unit bounces. The unit shall be subjected to this preconditioning for one minute. After bounce preconditioning, the unit shall not be repaired, alined, cleaned, or otherwise changed prior to subjection to inspection.

4.8 Service conditions tests. - Prior to service condition testing the equipment shall meet the performance specified in 3.8, except 3.8.5 (RFI).

4.8.1 Temperature test. - The equipment shall be subjected to the temperature cycling as specified on Drawing SM-B-432737.

4.8.2 Humidity (moisture-resistance) test. - The equipment shall be tested in accordance with Method 507, Procedure II, of MIL-STD-810. Prior to testing the equipment shall meet full specification performance for the measurements specified in 3.7.3. Measurements shall be made during the periods shown on the figure for the cycle. Do not move equipment from the humidity chamber for measurements, and do not leave power on after measurements have been completed. If any equipment fails to meet the specified performance during cycling, it shall be realigned or readjusted once and the test continued. No repair or replacement of parts shall be permitted. If the equipment fails to meet the specified performance during cycling, or fails subsequently, it does not pass the test. In addition, if the equipment fails to meet specified requirements after final conditioning and adjustment, it does not pass the test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:39Z
Check the source to verify that this is the current version before use.

4.8.3 Altitude (elevation) test. - The equipment shall be tested in accordance with the altitude test of Drawing SM-B-432737.

4.8.4 Vibration test. - The equipment shall be tested in accordance with the vibration test of Drawing SM-B-432737.

4.8.5 Shock test; bench-handling. - The equipment shall be tested in accordance with Method 516, Procedure V, of MIL-STD-810.

4.8.6 Shock test; drop. - The case shall be tested in accordance with the drop test portion of Drawing SM-B-432737.

4.8.7 Air-seal test. - The equipment shall be tested in accordance with the air seal test of Drawing SM-B-432737.

4.8.8 Fungus test. - The equipment shall be tested in accordance with Method 508, Procedure I of MIL-STD-810. Spraying shall include the parts and materials on and under the chassis, and the interior of the equipment case. The equipment shall remain in an open condition during the test. Operation is not required.

4.9 Performance testing. -

4.9.1 Power input. - During the performance tests detailed in SM-B-432731 (See 4.9.2) the input voltages and frequency shall be varied over the ranges indicated in 3.8.1. There shall be no degredation of the pulses over these ranges.

4.9.2 Antenna pulse simulator tests. - The simulated antenna pulses shall be tested for in accordance with SM-B-432731.

4.9.3 Testing of Block 2700. - Block 2700 shall be checked for conformance to 3.8.3.

4.9.3.1 Blown fuse indicator tests. - The fuses shall be removed from their holders. Each associated blown fuse indicator shall glow.

4.9.4 Position synchro tests. - The position synchros shall be tested in accordance with the drawings listed in 3.8.4.

4.9.5 Radio Interference suppression. - Radio interference tests shall be conducted in accordance with the following test categories of MIL-STD-461:

a. REO2 (14KHz to 1GHz)

b. A-CE1

c. RSO3 (14KHz to 1GHz)

d. A-CS1

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:39Z
Check the source to verify that this is the current version before use.

4.10 Testing with associated equipment. - When the subject equipment is being procured in conjunction with Computer CP-319/MPQ-4A and Indicator IP-375/MPQ-4A (Control-Indicator Group OA-1256/MPQ-4A) First Article testing shall be in accordance with SM-B-432737 and performance testing in accordance with SM-B-432713 (listed in SC-DL-432500).

4.11 Visual and mechanical inspection. - The equipment shall be examined for conformance with MIL-STD-454, Requirement 9.

4.12 Quality conformance inspection of preparation for delivery. - Preparation for delivery shall be inspected in accordance with Specification MIL-P-116 to determine conformance to the requirements of Section 5.

5. Preparation for Delivery. -

5.1. Preservation and packaging. - Preservation and packaging shall be Level A or C, as specified. (See 6.1)

5.1.1 Level A. -

5.1.1.1 Cleaning. - Each SM-154/MPQ-4A shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. - Each SM-154/MPQ-4A shall be dried in accordance with the applicable procedures of MIL-P-116.

5.1.1.3 Preservation application. - None required.

5.1.1.4 Unit packaging. - Unit packaging shall be in accordance with the methods prescribed in MIL-P-116, as specified herein.

5.1.1.4.1 Technical literature. - Each Technical literature shall be packaged Method IC-1.

5.1.1.4.2 SM-154/MPQ-4A. - SM-154/MPQ-4A shall be packaged Method III, as follows: Place the spare tubes, lamps and fuses within the cover of the simulator. Place one of the technical literature packaged as specified in 5.1.1.4.1 within the space provided in the cover. Close and secure the compartment covers. Place the simulator case cover on the simulator and secure the fastenings. Cushion the simulator case on all surfaces with cells and/or pads or both conforming to PPP-F-320, W5c weather resistant, designed and fabricated to protect all projections and absorb the impact in handling and transit. Place the cushioned simulator within a close fitting fiberboard box conforming to PPP-B-636, W5c weather resistant, place the technical literature packaged as specified in 5.1.1.4.1 in the box directly under the lid of the box. Box closure shall be as specified in the appendix of the box specification

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:39Z
Check the source to verify that this is the current version before use.

5.1.2 Level C. - SM-154/MPQ-4A shall be packaged in a manner that will afford adequate protection against corrosion, deterioration and damage during shipment from the supply source to the first receiving activity.

5.2 Packing. - Packing shall be Level A, B or C as specified (See 6.1)

5.2.1 Level A. - A quantity of SM-154/MPQ-4A, packaged as specified in 5.1 shall be packed within a close-fitting fiberboard box conforming to MIL-B-43291 or PPP-B-640, Class weather resistant, not to exceed the gross weight limitation of 200 pounds, closure shall be in accordance with the appendix of the applicable box specification. The shipping containers shall be reinforced by pressure-sensitive filament tape or none-metallic banding conforming to PPP-T-97, type IV or PPP-S-760, Type II, respectively, as specified in the appendix of the applicable box specification.

5.2.2 Level B. - SM-154/MPQ-4A, packaged as specified in 5.1 shall be packed in accordance with 5.2.1.

5.2.3 Level C. - SM-154/MPQ-4A, packaged as specified in 5.1 shall be packed in shipping containers in a manner that will afford adequate protection against damage to the package and its contents during shipment from the supply source to the first receiving activity, shipping containers shall comply with the rules and regulations of the command carrier as applicable to the mode of transportation.

5.3 Marking. - In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. - Procurement documents should specify the following:

a. Title, number, and date of this specification and any amendment thereto.

b. Type required.

c. Level of packaging and level of packing required for shipment. (Level A or Level C).

d. The specific paragraphs of Section 5 which are applicable to the particular procurement.

e. First Article inspection:

(1) Two sample units of each item cited in Section 1 are generally required so that lengthy environmental tests can be completed on one sample unit while complete performance measurements can be made on the second sample unit. (See 3.2)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:39Z
Check the source to verify that this is the current version before use.

(2) First Article pack(s) as follows:

Makeup of pack(s).

Number of each kind of pack to be submitted.

Inspection to be performed thereon.

f. Marking and shipping of samples.

g. Place of final inspection.

h. Technical literature required. (See 3.9)

i. Quantity of tools and running spare parts required. (See 3.9)

6.2 Group C inspection. - Approval to ship may be withheld, at the discretion of the Government, pending the decision from the contracting officer on the adequacy of corrective action. (See 4.5.3.2.)

6.3 Location of air-seal test. - It is desirable that the air-seal test (4.8.7) be performed at a location that will minimize handling (which might cause damage to the equipment) after this inspection is completed. Any preparation for shipment which would require breaking of the equipment seal should be accomplished prior to the air-seal test so that the seal may remain intact thereafter. It is recommended that the entire lot (including all previously inspected sample units) be sampled and inspected immediately prior to packaging.

6.4 Nomenclature. - The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example: SM-154W/MrQ-4A. The contractor should apply for nomenclature in accordance with the applicable clause in the contract. (See 1.1)

6.5 Verification inspection. - Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

Custodian:

Army-EL

Preparing activity:

Army-EL

Project No. 6625-A403

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:39Z
Check the source to verify that this is the current version before use.

FOLD

Department of the Army
Headquarters
U.S.Army Electronics Command
Fort Monmouth, New Jersey 07703

POSTAGE AND FEES PAID

OFFICIAL BUSINESS

Commanding General
U.S.Army Electronics Command
ATTN: AMSEL-TD-SS
Fort Monmouth, New Jersey 07703

FOLD

To detach this form, cut along this line

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:39Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

MIL-S-55226A(EL), SIMULATOR, ANTENNA POSITION SM-154( )/MPQ-4A

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

ESC-FM 1068-68

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:39Z
Check the source to verify that this is the current version before use.